

RECORD AUTHORITY

Post-test inspection and calibration review

Signed quality record for the same test article AQL-P3-017 after completion of AQL-DVTR-118. This dossier is authoritative for destructive internal inspection, photographed hardware observations and certificate validity.

Table 1. Record control

Field	Value
Document type	Quality inspection and qualification record
Inspector	R. Singh - signed 2026-07-21
Test article	AQL-P3-017
Related report	AQL-DVTR-118 Rev A

Interpretation. A later direct internal inspection can contradict an earlier external or in-run observation when the two records address the same acceptance condition.

VISUAL MARKING AUDIT

VI-02 - Service-disconnect marking photograph

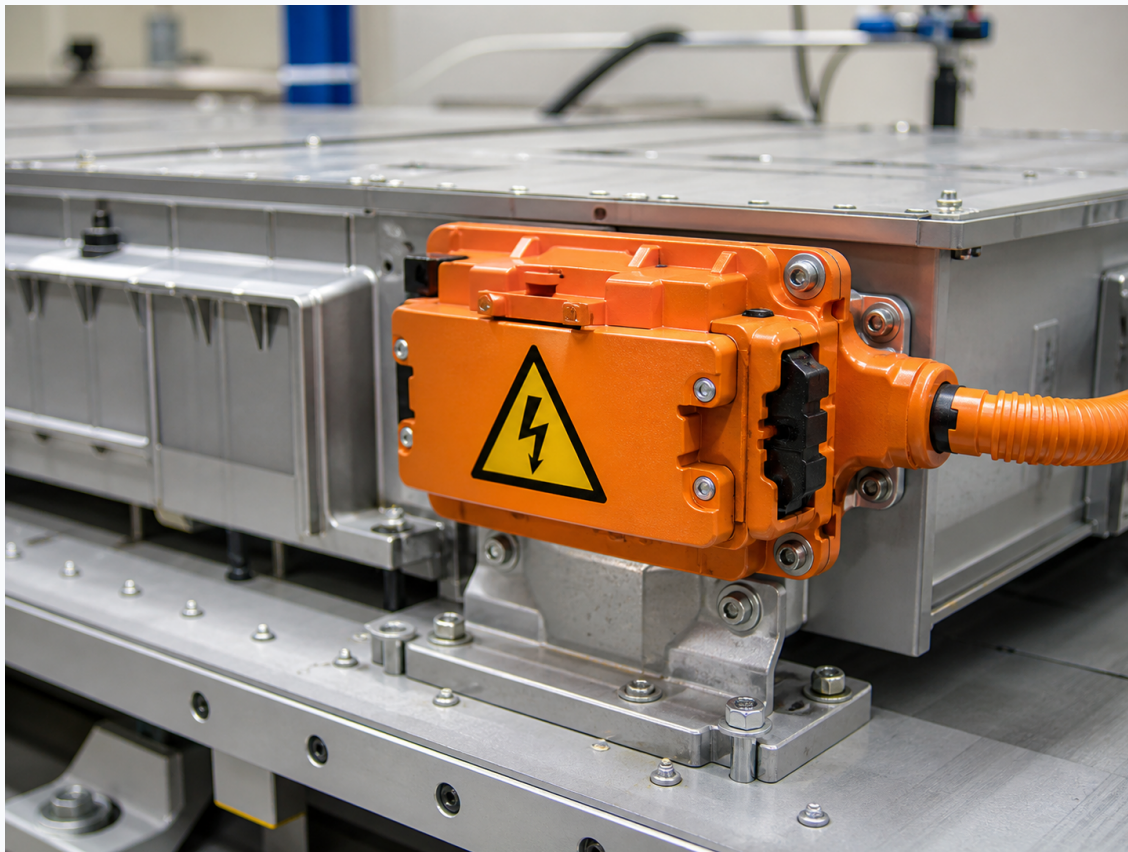


Figure 1. Photograph supplied by the inspection team: orange service-disconnect housing with yellow triangular black-lightning hazard symbol.

Traceability limitation. The photograph contains no visible unit serial or station identifier. Only one cover is shown, while the requirement applies to every service-disconnect cover. Population coverage and identity of the photographed cover cannot be confirmed.

Inspection disposition: INCONCLUSIVE for full requirement scope. The visual design appears consistent, but this record cannot prove that all covers are marked before access.

COOLANT POST-TEST INSPECTION

FI-03 - Coolant circuit close inspection

Table 2. Post-hold inspection findings

Location	Method	Observation	Disposition
Quick disconnect QD-1	White-cloth wipe	Dry	No finding
Quick disconnect QD-2	White-cloth wipe	Blue coolant residue; collected volume 0.4 mL	VISIBLE LEAKAGE
Manifold seam	UV lamp	Dry	No finding

Conflict with test report. The chamber-window observation in AQL-DVTR-118 recorded 0.00 mL during the pressure hold. The later direct QD-2 inspection found visible coolant residue. For a no-visible-leakage condition, these are conflicting observations and require review.

J17 DESTRUCTIVE INGRESS INSPECTION

IP-04 - Connector J17 rear-boot inspection

Table 3. Internal inspection after immersion

Step	Observation
Rear boot opened	Liquid droplets present behind boot seal
Collected liquid	0.6 mL, conductivity consistent with immersion bath
Contact pins	No corrosion at inspection time
Conclusion	Moisture ingress confirmed; IP67 no-ingress condition failed

J17 connector destructive inspection cross-section

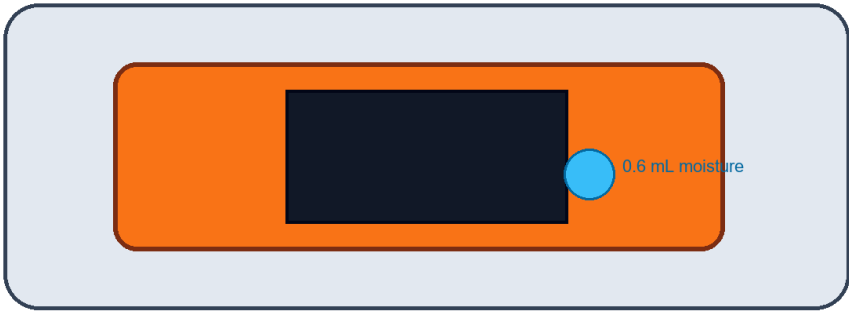


Figure 2. Blue marker indicates collected liquid behind the rear boot after opening.

Figure 2. Inspection cross-section showing the location of collected moisture behind the rear boot.

Direct contradiction. AQL-DVTR-118 records an external visual PASS before disassembly. This destructive inspection directly observes internal moisture and therefore contradicts the no-ingress acceptance condition.

CALIBRATION CERTIFICATE REVIEW

CAL-05 - Instrument certificate validation

Table 4. Certificate status on test date 2026-07-18

Serial	Certificate issue	Valid through	Age/status on test date
DMM-22	2026-02-12	2027-02-11	5 months - VALID
DAQ-14	2026-01-29	2027-01-28	6 months - VALID
HI-07	2025-05-03	2026-05-02	Expired 77 days before test - INVALID

Finding CAL-05-03. HI-07 produced the acceptance isolation-resistance value but its calibration was not valid within 12 months on the test date. Instrument identity is traceable; calibration validity is not compliant.